ROMA TRE
UNIVERSITÀ DEGLI STUDI

Information Visualization

Infovis on the Web

Graphics

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Credits

- Parts of this matherial is inspired by
  - http://www.teaching-materials.org/
  - https://developer.mozilla.org/en-US/docs/Web/SVG/Tutorial

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Raster Vs Vector graphics

- Raster graphics
  - based on pixels (picture element) arranged in a grid (= bitmaps)
    - a pixel represents the smallest unit of a video image that has specific RGBA values
  - lower abstraction level
- Vector graphics
  - based on geometrical primitives such as points, lines, curves and polygon(s)
    - mathematical expressions
  - more abstract level

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Raster properties

- Resolution
  - number of pixels to represent the image
- Color depth
  - number of bits to color a single pixel
  - typically 24-bits for RGB
    - eight bits for each component
    - $2^8 = 256$ shades per component
    - $256^3 \sim 16$ million colors
  - humans distinguish ~ 10 mil colors [Judd ‘75]
  - often 8 more bits are used for opacity

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Vector graphic properties

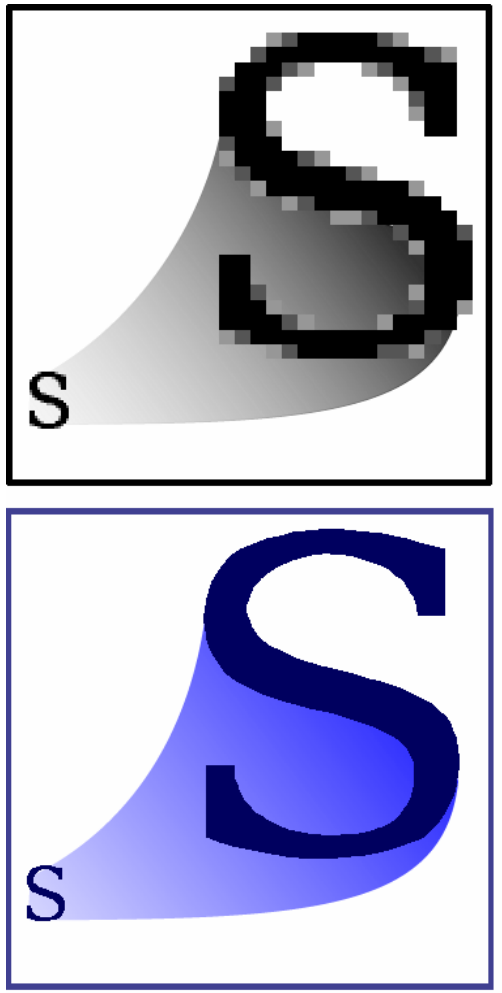
- Vector graphics does not keep track of each pixel
  - it only considers the mathematical information that is able to generate the picture
- It is based on shapes (ultimately on paths)
  - paths are defined by starting and ending points, together with other control points, curves, and angles along the way
  - paths are associated with styles
    - fill color, stroke-width, etc.

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Raster vs Vector: pros and cons

- Bitmaps
  - fine-grained control up to single pixels
  - best for full-color images, like photographs
  - increasing resolution or color depth affects their size
  - do not resize well

- Vector graphics
  - resolution independency
  - resize with little or no loss
  - no extremely complex images
  - time and talent needed to create it



[Image from Wikipedia.org]

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HTML5 Canvas and SVG

- HTML5 Canvas element
  - “provides scripts with a resolution-dependent bitmap canvas, which can be used for rendering graphs, game graphics, or other visual images on the fly” [W3C]
- SVG (Scalable Vector Graphics)
  - “is an XML-based format for describing 2D vector graphics” [W3C]

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High level summary

HTML5 Canvas	SVG
Pixel based	Shape based
Single HTML element	Multiple graphical elements, which become part of the DOM
Modified through script only	Modified through script and CSS
Interaction is granular (x,y)	Interaction is abstracted (rect, path,…)
Performance is better with smaller surface, a larger number of objects (>10K), or both	Performance is better with larger surface, a smaller number of objects (<10K), or both
Stateless (next slide)	Statefull (next slide)

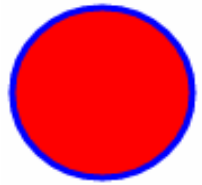
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Stateless and stateful paradigms

- Canvas is stateless
  - drawing primitives change the color of a pixel
    - irrespectively of previous drawing instructions
- SVG is stateful
  - drawing primitives change the properties of an object
    - which has been defined earlier
  - this implies easier animation and interaction

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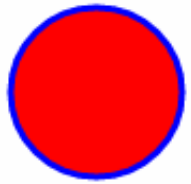
A circle in Canvas



```
<canvas id="myCanvas" width="100" height="100"></canvas>
```

```
<script>
function drawCircle(radius) {
  var canvas = document.getElementById('myCanvas');
  var context = canvas.getContext('2d');
  var centerX = canvas.width / 2;
  var centerY = canvas.height / 2;
  context.clearRect(0, 0, canvas.width, canvas.height);
  context.beginPath();
  context.arc(centerX, centerY, radius, 0, 2 * Math.PI);
  context.fillStyle = 'red';
  context.fill();
  context.lineWidth = 3;
  context.strokeStyle = 'blue';
  context.stroke();
}
drawCircle(40);
</script>
```


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A cycle in SVG 

```
<svg height="100"
     width="100“
     xmlns="http://www.w3.org/2000/svg">
  <circle cx="50"
          cy="50"
          r="40"
          stroke="blue"
          stroke-width="3"
          fill="red">
  </circle>
</svg>
```

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A simple animation with Canvas

```
<script>
var i = 10;
setInterval(
  function(){
        drawCircle(i);
        if(i < 40) i++;
    }, 25);
</script>
```



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A simple animation with SVG

```
<script>
var i = 10;
setInterval(
  function(){
        document.getElementById("myCircle")
                .setAttribute("r", i);
        if(i < 40) i++;
    }, 25);
</script>
```



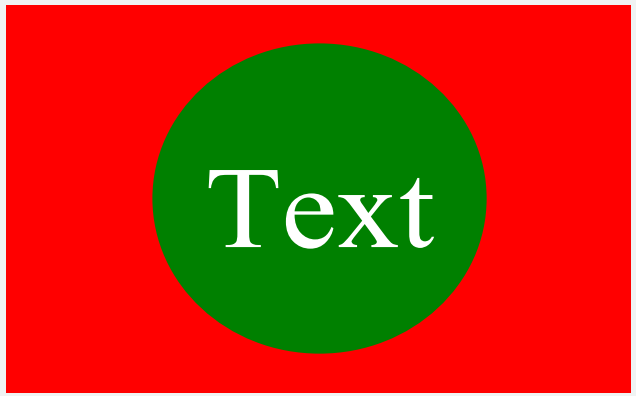
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SVG: basic example

- Objects are rendered in the order in which they are listed

```
<svg width="300"
     height="200"
     xmlns="http://www.w3.org/2000/svg">

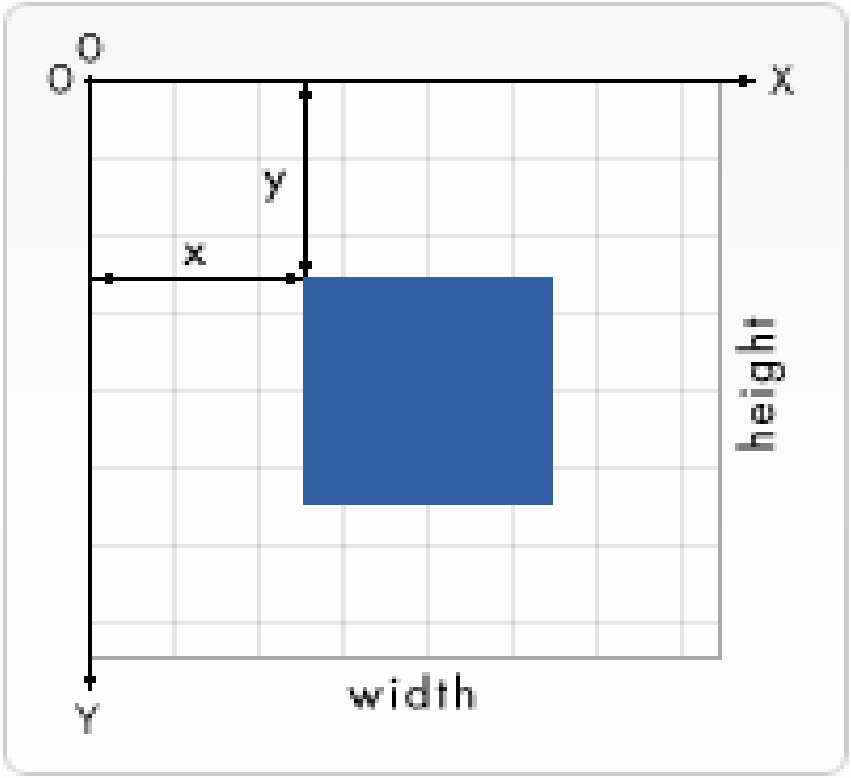
<rect width="100%" height="100%" fill="red" />
<circle cx="150" cy="100" r="80" fill="green" />
<text x="150"
      y="125"
      font-size="60"
      text-anchor="middle"
      fill="white"> Text
</text>
</svg>
```

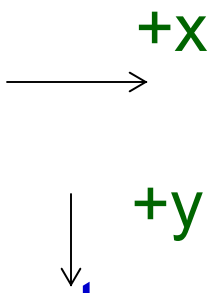


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Positions

- SVG uses a grid system
- The top left corner of the document is (0,0)
- Positions are measured in **pixels** with
  - positive x direction to the right
  - positive y direction to the bottom
- One pixel in SVG maps to one pixel on the output device





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Basic shapes

```
<svg width="200" height="250" version="1.1" xmlns="http://www.w3.org/2000/svg">
   <rect x="10" y="10" width="30" height="30" stroke="black" fill="transparent"
      stroke-width="5"/>
   <rect x="60" y="10" rx="10" ry="10" width="30" height="30" stroke="black"
      fill="transparent" stroke-width="5"/>
   <circle cx="25" cy="75" r="20" stroke="red" fill="transparent"
        stroke-width="5"/>
   <ellipse cx="75" cy="75" rx="20" ry="5" stroke="red" fill="transparent"
         stroke-width="5"/>
   <line x1="10" x2="50" y1="110" y2="150" stroke="orange" fill="transparent"
      stroke-width="5"/>
   <polyline points="60 110, 65 120, 70 115, 75 130, 80 125, 85 140, 90 135,
         95 150" stroke="orange" fill="transparent" stroke-width="5"/>
   <polygon points="50 160 55 180 70 180 60 190 65 205 50 195 35 205 40 190 30
         180 45 180" stroke="green" fill="transparent" stroke-width="5"/>

   <path d="M20,230 Q40,205 50,230 T90,230" fill="none" stroke="blue"
      stroke-width="5"/>

</svg>
```



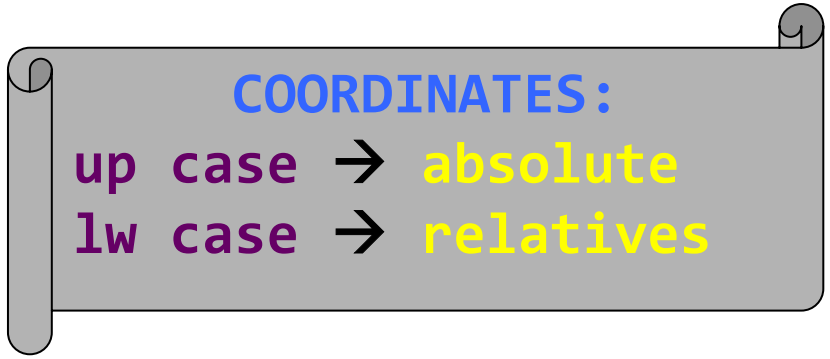
SVG Path Commands

[MDNP, https://developer.mozilla.org/en-US/docs/Web/SVG/Tutorial/Paths]

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Paths: Line Commands

- M x y: “move to”
  - it does not draw anything
  - it moves the cursor
- L x y: “line to”
  - it draws a line from the current position to (x, y)
- H x or V y: “horizontal to” or “vertical to”
  - it draws horizontal and vertical lines
- Z: “close path”
  - it draws a line from the current position back to the first point

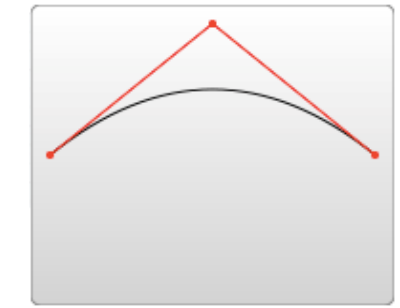


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Paths: quadratic bezier curves (Q)

- Q x1 y1, x y
- It requires one control point
  - which determines the slope of the curve at both the start point and at the end point
- It is simpler than the cubic one

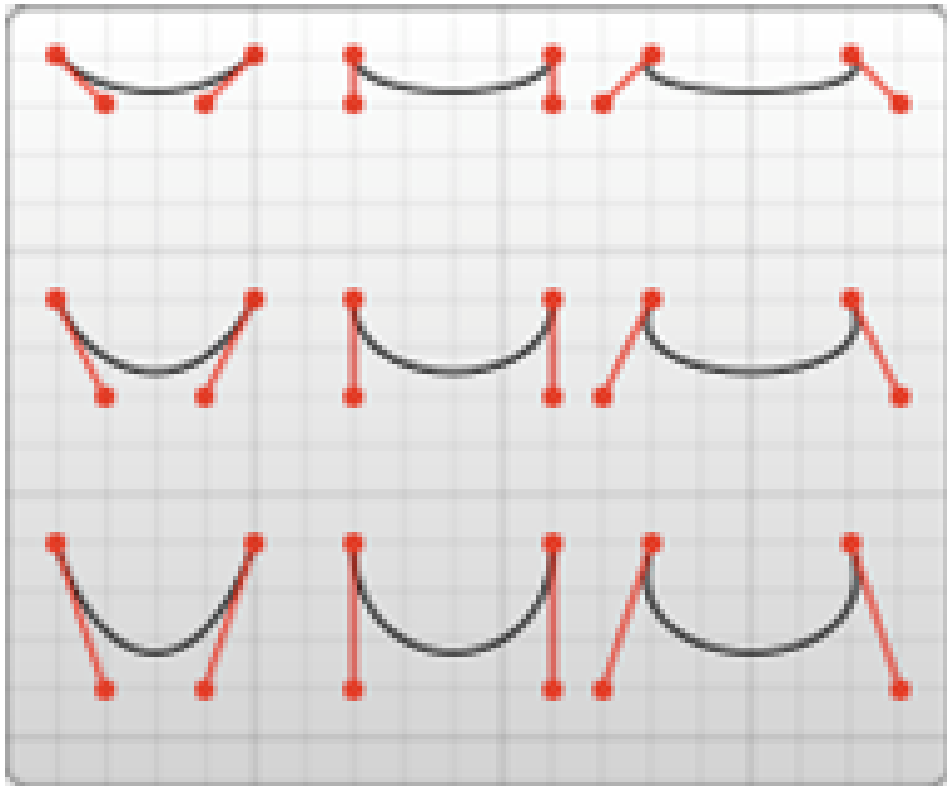
```
<path d="M10 80 Q 95 10, 180 80"
stroke="black"
fill="transparent"/>
```



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Paths: cubic bezier curves (C)

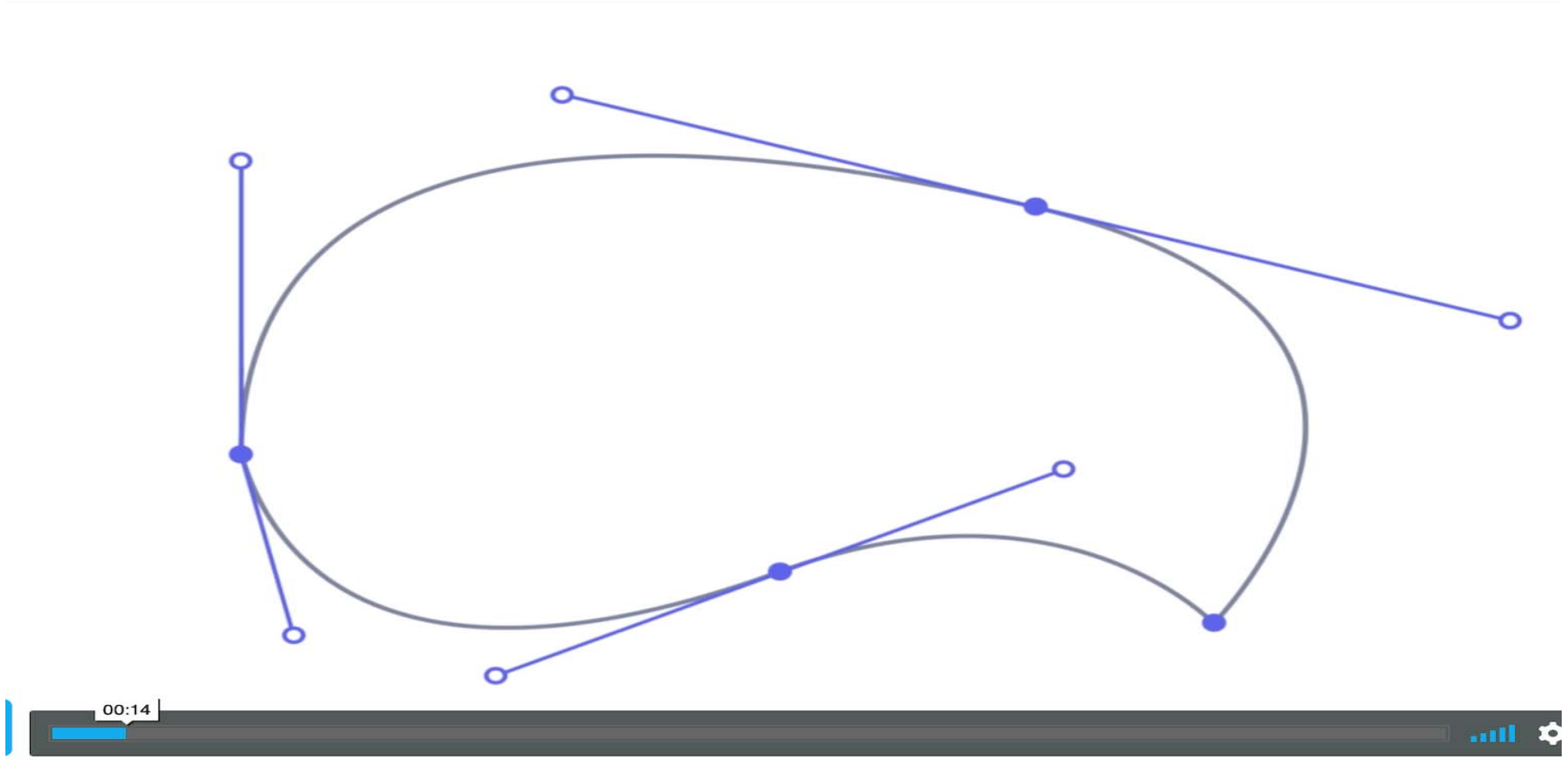
- C x1 y1, x2 y2, x y
- It requires two control points
  - which determines the slope of the curve at both the start point and at the end point



```
<path d="M10 10 C 20 20, 40 20, 50 10"
stroke="black” fill="transparent"/>
```


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Cubic Bezier Curves
Under the Hood



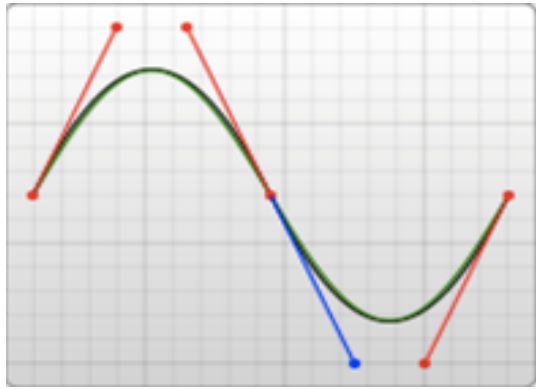
https://vimeo.com/106757336

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Paths: compose curves

- S x2 y2, x y
- You can string together several Bezier curves to create extended smooth shapes
  - the control point on one side of a point will be a reflection of the control point used on the other side to keep the slope constant
  - in this case, you can use a shortcut version of the cubic Bezier

```
<path d="M10 80 C 40 10, 65 10, 95 80 S 150 150,
180 80” stroke="black" fill="transparent"/>
```



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Elliptical Arcs (1/9)

- Arcs are sections of circles or ellipses
  - A rx ry, x-axis-rotation, l-flag, sweep-flag, x y
    - rx = x-radius
    - ry = y-radius
    - x-axis-rotation = clockwise rotation of the ellipse (degrees)
    - l-flag = 0/1

      0 means “use one of the two smaller arcs”

      1 means “use one of the two larger arcs”
    - sweep-flag = 0/1

      0 means “counter-clockwise”

      1 means “clockwise”
    - x y = coordinates of the final point

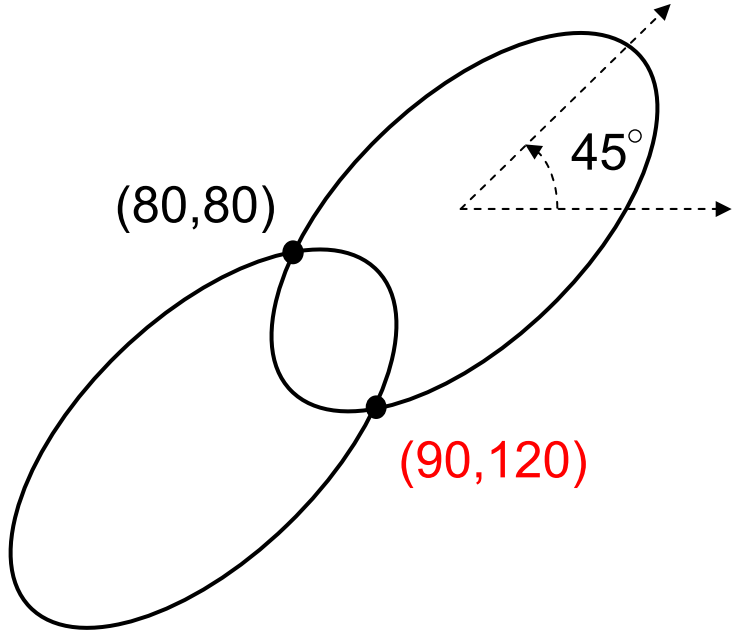
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Elliptical Arcs (2/9)

- Given rx, ry, and x-axis-rotation there are two ellipses that can connect the starting and ending points

```
<path d="M80 80 A 50 20, 45, ..., 90 120” stroke="black"/>
```

- Along each one of these two ellipses there are two possible arcs that you can use
  - overall, you have four different arcs

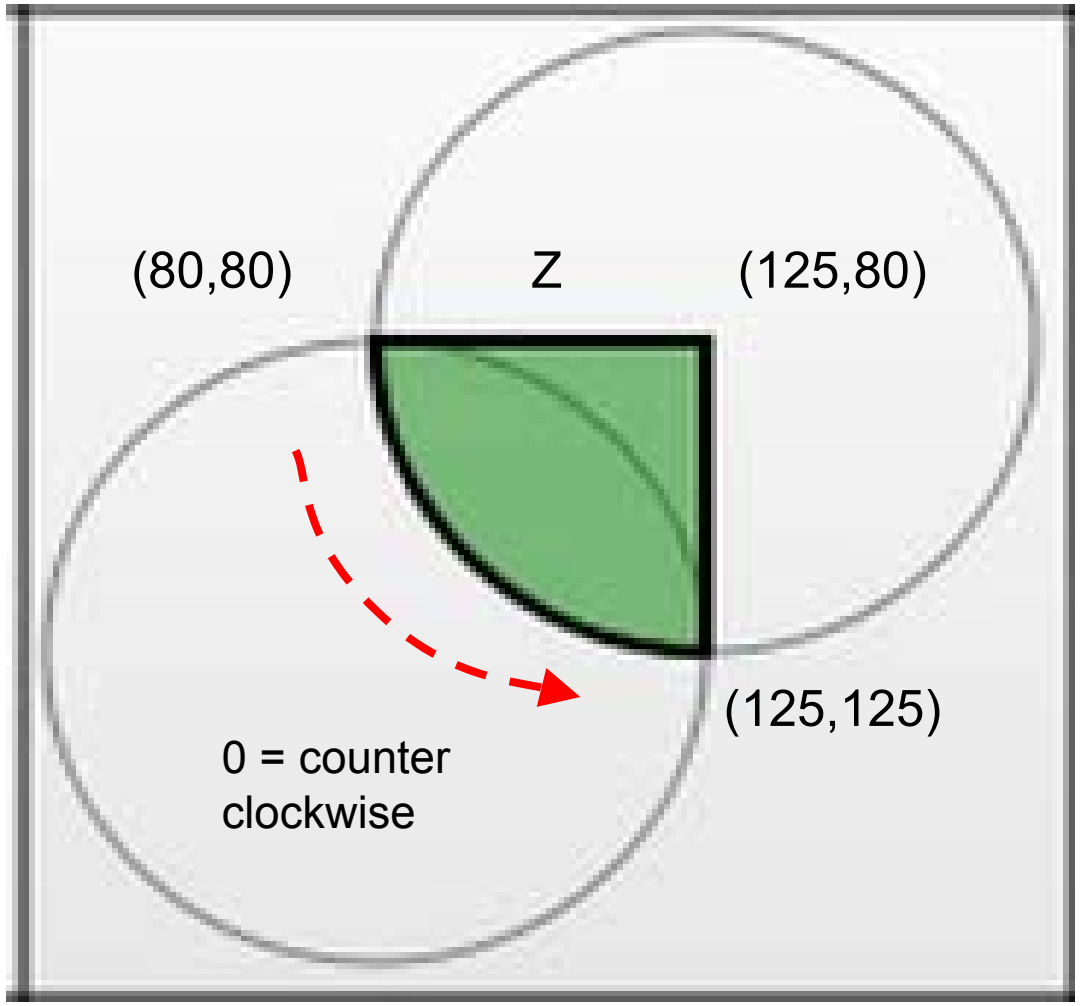


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Elliptical Arcs (3/9)

```
<path d="M80 80 A 45 45, 0, 0, 0, 125 125 L 125 80 Z"
fill="green"/>
```

- l-flag = 0/1
  - 0 means “use one of the two smaller arcs”
  - 1 means “use one of the two larger arcs”
- sweep-flag = 0/1
  - 0 means “counter-clockwise”
  - 1 means “clockwise”

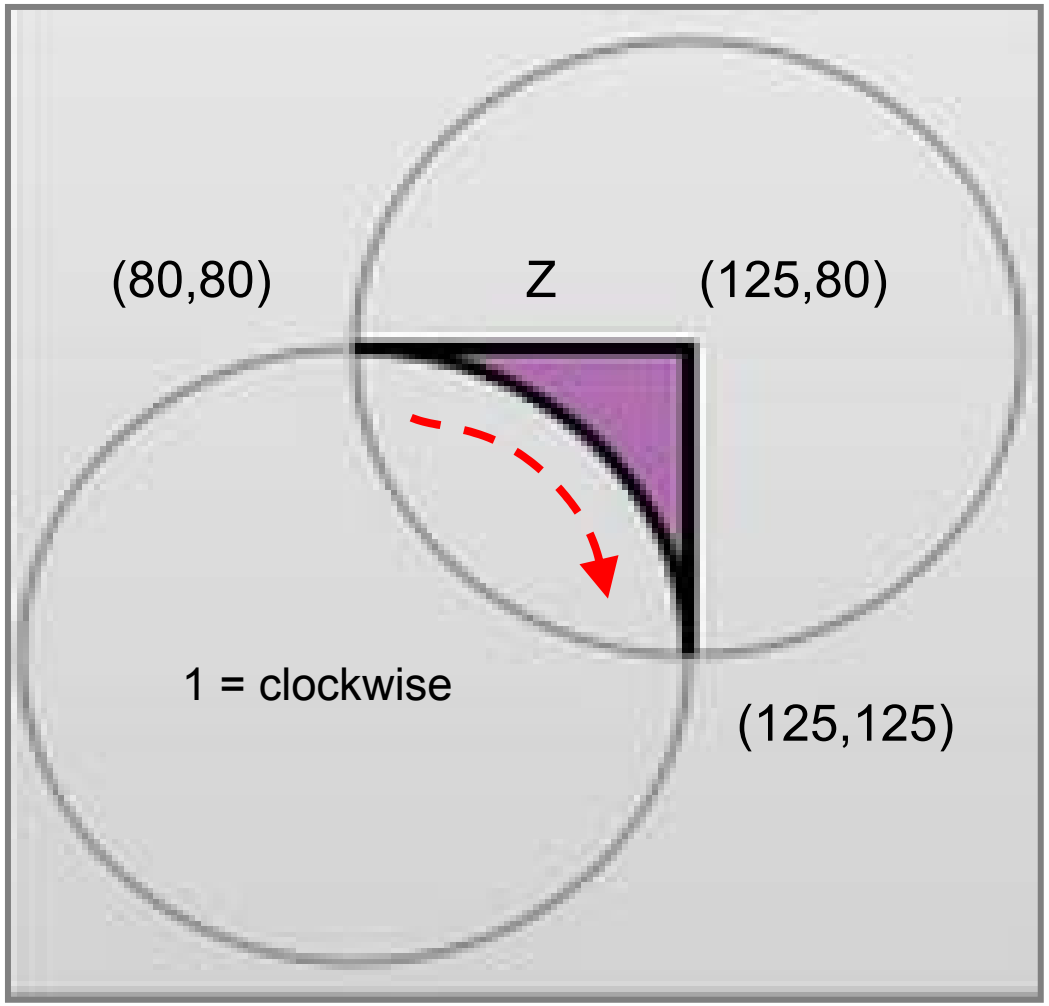


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Elliptical Arcs (4/9)

```
<path d="M80 80 A 45 45, 0, 0, 1, 125 125 L 125 80 Z"
fill="purple"/>
```

- l-flag = 0/1
  - 0 means “use one of the two smaller arcs”
  - 1 means “use one of the two larger arcs”
- sweep-flag = 0/1
  - 0 means “counter-clockwise”
  - 1 means “clockwise”

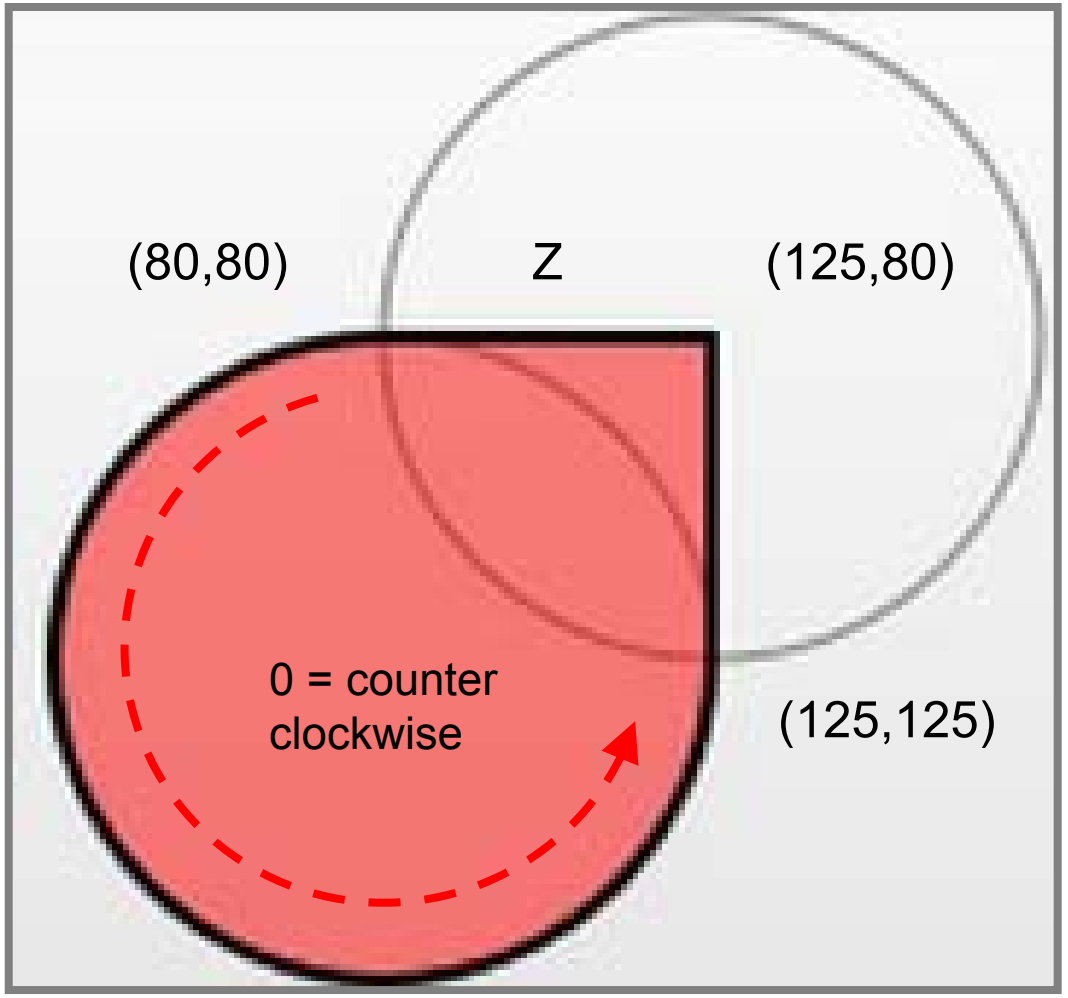


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Elliptical Arcs (5/9)

```
<path d="M80 80 A 45 45, 0,
1, 0, 125 125 L 125 80 Z"
fill="red"/>
```

- l-flag = 0/1
  - 0 means “use one of the two smaller arcs”
  - 1 means “use one of the two larger arcs”
- sweep-flag = 0/1
  - 0 means “counter-clockwise”
  - 1 means “clockwise”

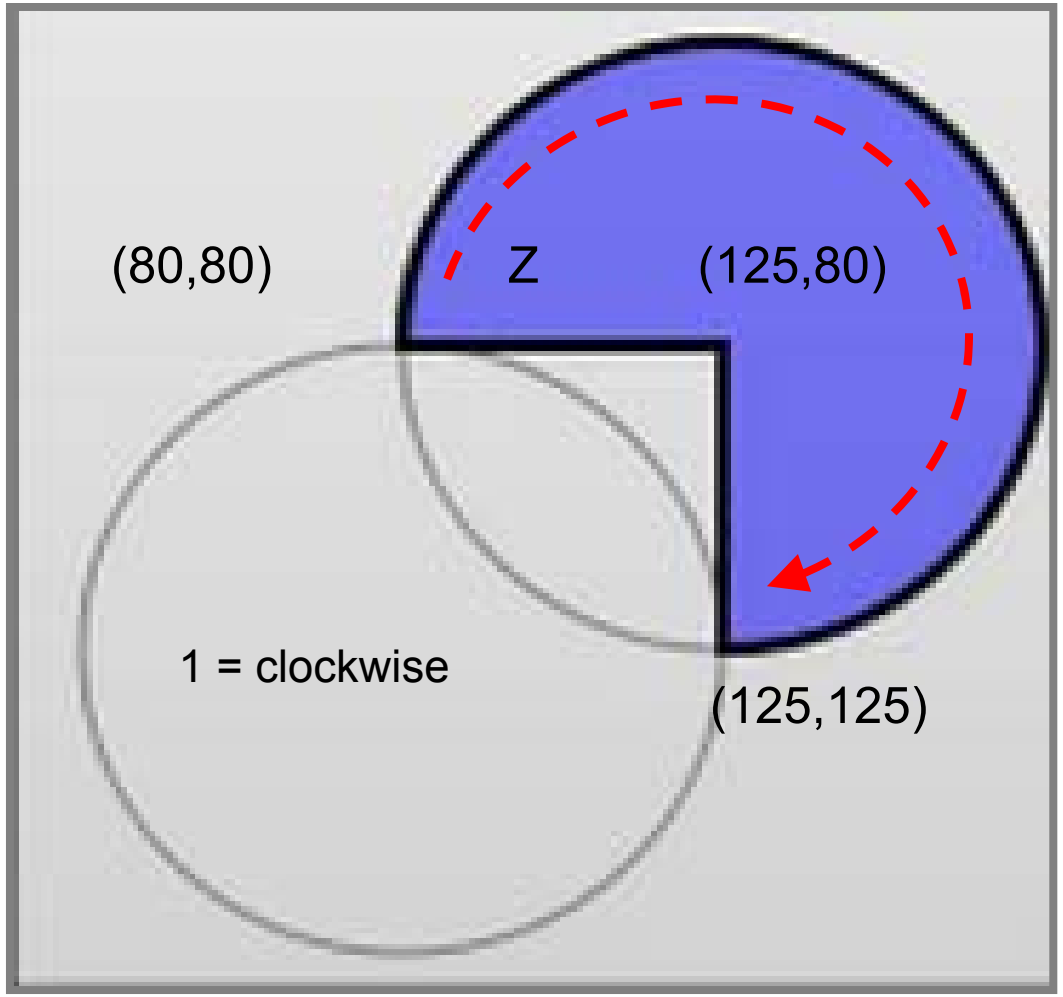


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Elliptical Arcs (6/9)

```
<path d="M80 80 A 45 45, 0,
1, 1, 125 125 L 125 80 Z"
fill="blue"/>
```

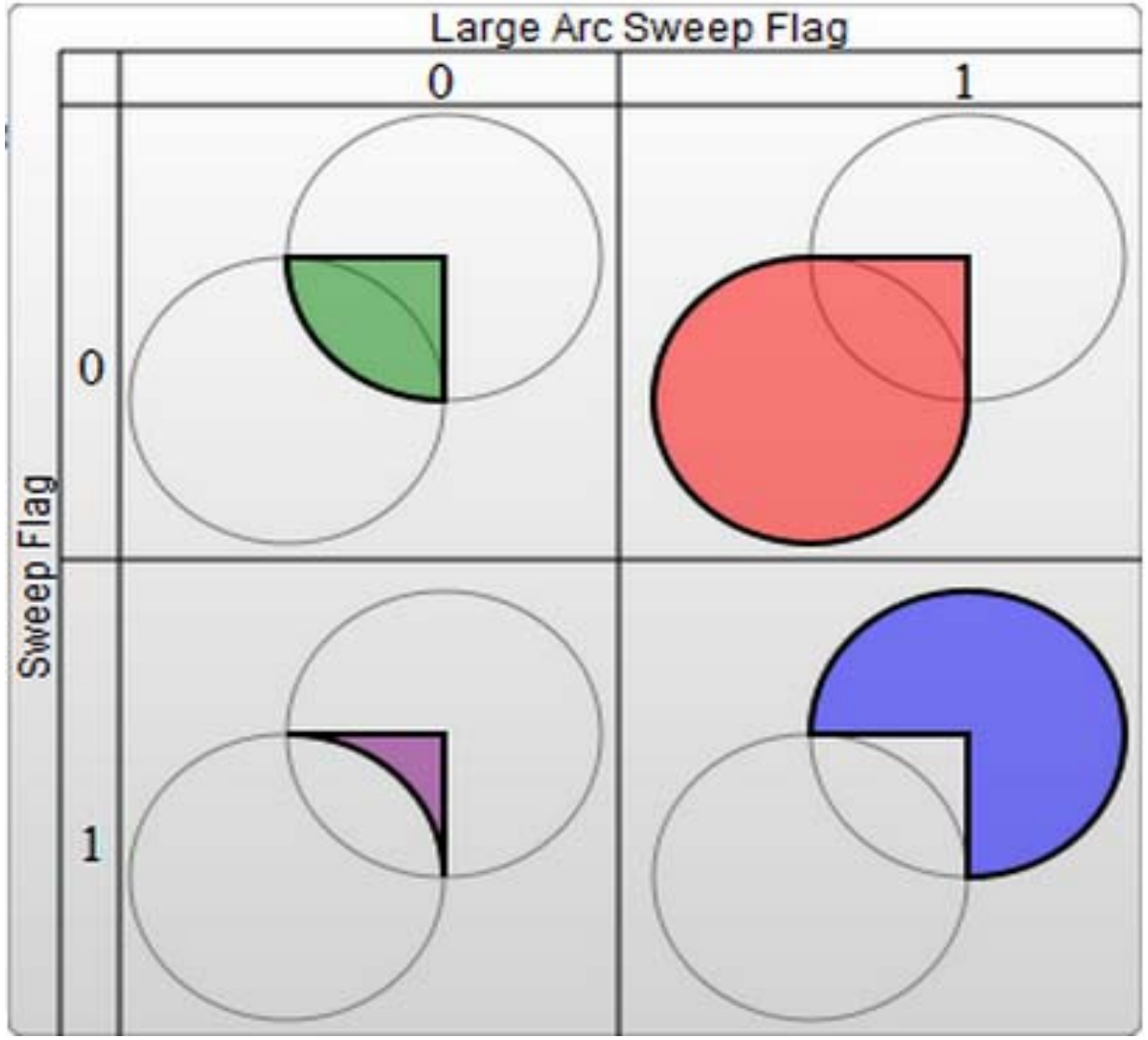
- l-flag = 0/1
  - 0 means “use one of the two smaller arcs”
  - 1 means “use one of the two larger arcs”
- sweep-flag = 0/1
  - 0 means “counter-clockwise”
  - 1 means “clockwise”



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Elliptical Arcs (7/9)

```
<path d="M80 80 A 45 45, 0,
0, 0, 125 125 L 125 80 Z"
fill="green"/>
<path d="M230 80 A 45 45, 0,
1, 0, 275 125 L 275 80 Z"
fill="red"/>
<path d="M80 230 A 45 45, 0,
0, 1, 125 275 L 125 230 Z"
fill="purple"/>
<path d="M230 230 A 45 45, 0,
1, 1, 275 275 L 275 230 Z"
fill="blue"/>
```

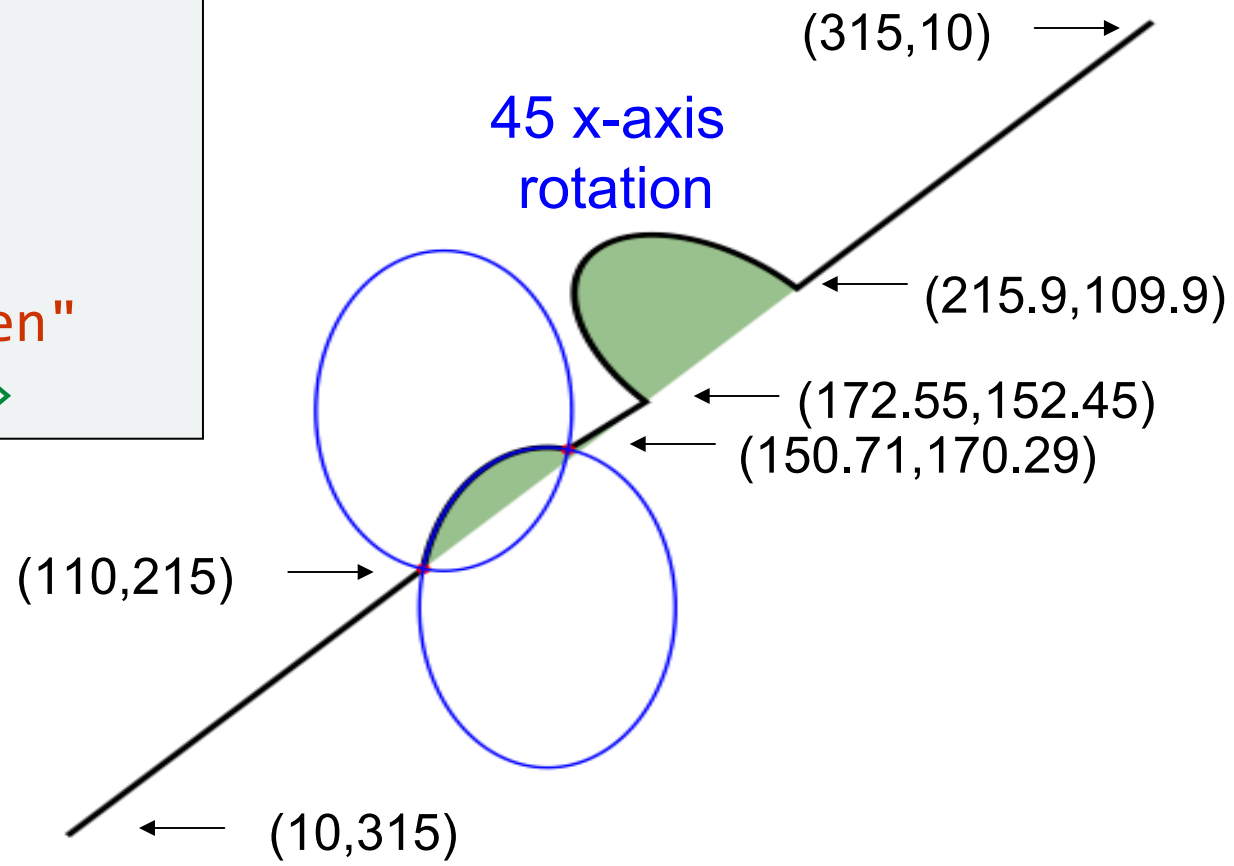


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Elliptical Arcs (8/9)

- Example

```
<path d="M10 315
  L 110 215
  A 36 60, 0, 0, 1, 150.71 170.29
  L 172.55 152.45
  A 30 50, -45, 0, 1, 215.1 109.9
  L 315 10" stroke="black" fill="green"
stroke-width="2" fill-opacity="0.5"/>
```

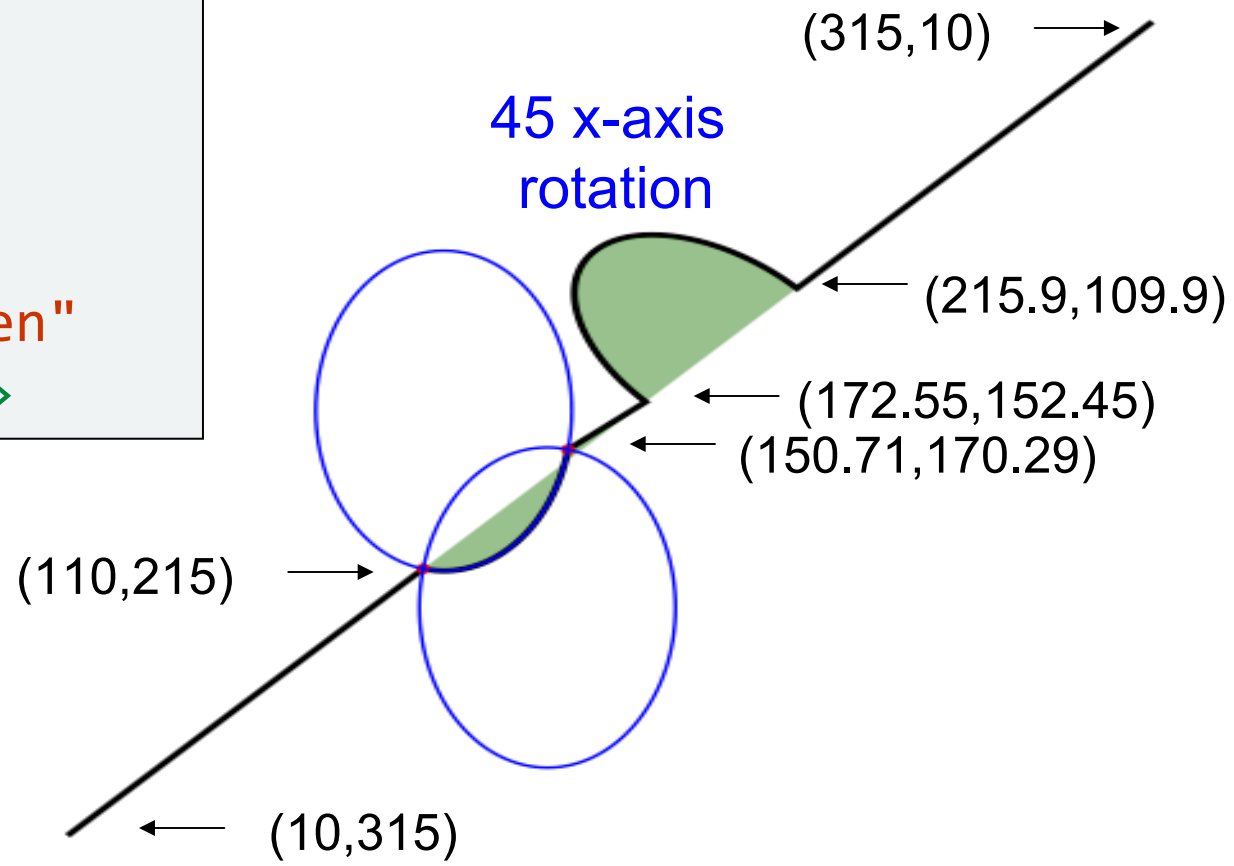


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Elliptical Arcs (9/9)

- Example

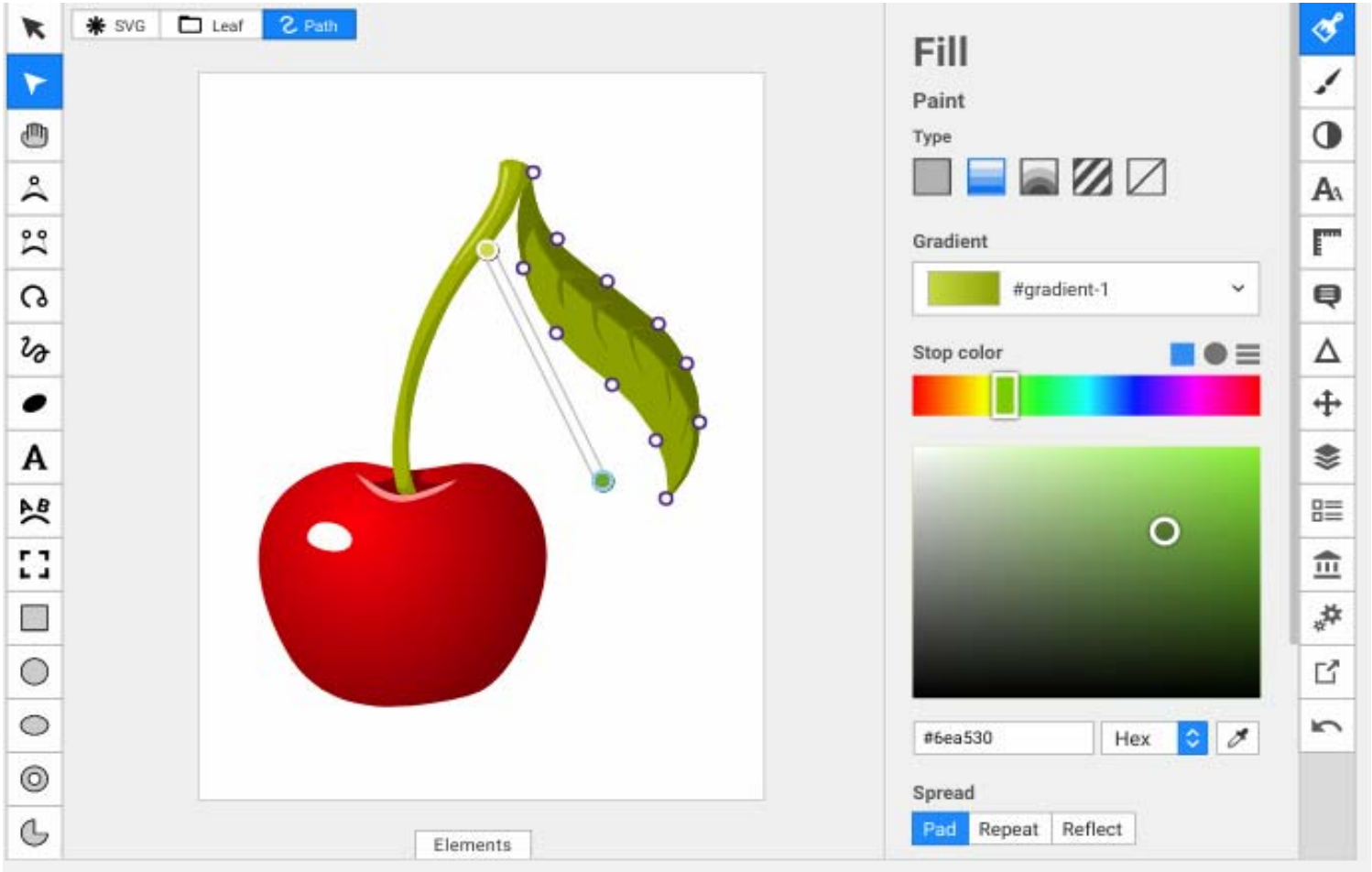
```
<path d="M10 315
  L 110 215
  A 36 60, 0, 0, 0, 150.71 170.29
  L 172.55 152.45
  A 30 50, -45, 0, 1, 215.1 109.9
  L 315 10" stroke="black" fill="green"
stroke-width="2" fill-opacity="0.5"/>
```



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SVG editors

- https://boxy-svg.com/



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Basic transformations: translations

- Translations
  - it may be necessary to move an element around
  - you can use the translate() transformation

```
<rect x="0" y="0" width="10" height="10"
      transform="translate(30,40)" />
```

  - the example will render a rectangle, translated to the point (30,40) instead of (0,0)

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Basic transformations: rotations

- Rotations
  - rotating an element is quite a common task
  - use the rotate() transformation for this

```
<rect x="20" y="20" width="20" height="20"
      transform="rotate(45)" />
```

  - this example shows a square that is rotated by 45 degrees
    - the value for rotate() is given in degrees

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Basic transformations: scalings

- Scalings
  - scale() changes the size of an element
  - it takes two numbers, evaluated as ratio by which to scale on x and y
  - 0.5 shrinks by 50%

```
<rect x="20" y="20" width="20" height="20"
      transform="scale(0.8,0.8)" />
```

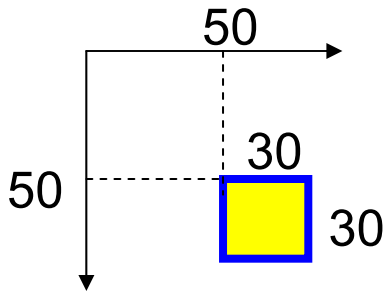
  - if the second number is omitted, it is assumed to be equal to the first

```
<rect x="20" y="20" width="20" height="20"
      transform="scale(0.8)" />
```

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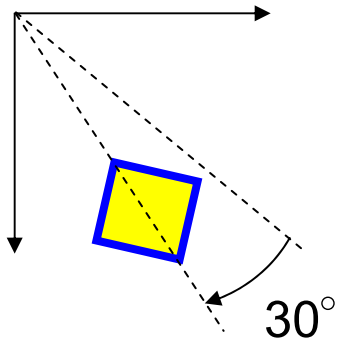
Concatenation of transformations

- A yellow rectangle




```
<rect x="50"
      y="50"
      width="30"
      height="30"
      stroke="blue"
      stroke-width="3"
      fill="yellow"
/>
```

- A rotated rectangle

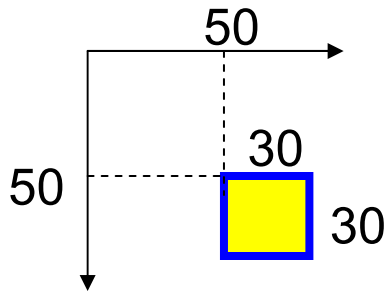


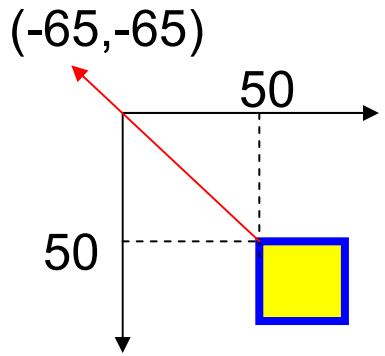

```
<rect x="50"
      y="50"
      width="30"
      height="30"
      stroke="blue"
      stroke-width="3"
      fill="yellow"
      transform="rotate(30)"
/>
```


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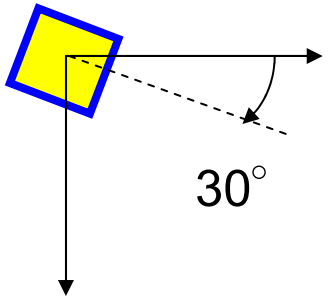
Concatenation of transformations

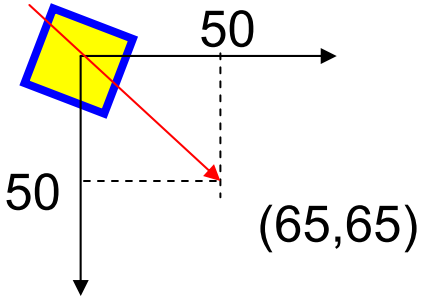
- transformations are applied in reverse order

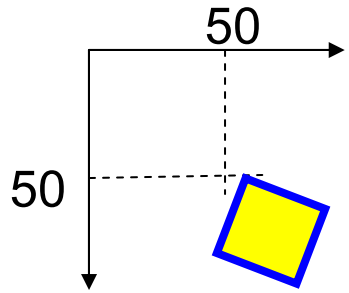





```
<rect x="50"
      y="50"
      width="30"
      height="30"
      stroke="blue"
      stroke-width="3"
      fill="yellow"
      transform="translate(65,65),
                 rotate(30),
                 translate(-65,-65)"
/>
```







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Bibliography and links

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- [Judd ‘75] Deane B. Judd, “Color in business, science and industry”. Wiley-Interscience, 3rd ed., 1975
- [W3C] http://www.w3.org/TR/SVG/intro.html
- [TM] http://www.teaching-materials.org/
- [MDN] https://developer.mozilla.org/en-US/docs/Web/SVG/Tutorial
- [MDNP] https://developer.mozilla.org/en-US/docs/Web/SVG/Tutorial/Paths

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